WORKED SOLUTION

# STEP III 2010, Question 1

Pure • Mean and Variance

**STEP III 2010, Question 1**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q1

## The key idea

$A$ and $B$ are the mean and variance of the first $n$ numbers; $C$ and $D$ are the mean and variance once one more number is added. Variance is easiest to handle in the form "mean of the squares minus the square of the mean", which is what part (ii) gives us. With that, $D$ comes out as a quadratic in $x_{n+1}$, and both inequalities are statements about that quadratic.

## Solution

Throughout, write $x = x_{n+1}$ for brevity.

### Part (i)

By definition $\sum_{k=1}^{n} x_k = nA$, so

$$C = \frac{1}{n+1}\left(\sum_{k=1}^{n} x_k + x\right) = \frac{nA + x}{n+1}.$$

### Part (ii)

Expand the square and use $\sum_{k=1}^{n} x_k = nA$ and $\sum_{k=1}^{n} 1 = n$:

$$\begin{aligned} B &= \frac{1}{n}\sum_{k=1}^{n}(x_k^2 - 2Ax_k + A^2) = \frac{1}{n}\sum_{k=1}^{n} x_k^2 - \frac{2A}{n}\cdot nA + \frac{1}{n}\cdot nA^2 \\ &= \frac{1}{n}\sum_{k=1}^{n} x_k^2 - A^2, \end{aligned}$$

as required.

**Part (iii)**

Part (ii) holds for any list of numbers, so applying it to the $n+1$ numbers $x_1, \ldots, x_{n+1}$ (with mean $C$) gives

$$D = \frac{1}{n+1}\sum_{k=1}^{n+1} x_k^2 - C^2.$$

From part (ii) again, $\sum_{k=1}^{n} x_k^2 = n(B+A^2)$, so $\sum_{k=1}^{n+1} x_k^2 = n(B+A^2) + x^2$. Using part (i) for $C$,

$$D = \frac{nB + nA^2 + x^2}{n+1} - \frac{(nA+x)^2}{(n+1)^2}.$$

Multiply through by $(n+1)^2$ and expand:

$$\begin{aligned}(n+1)^2 D &= (n+1)(nB + nA^2 + x^2) - (n^2A^2 + 2nAx + x^2)\\ &= n(n+1)B + (n^2 + n - n^2)A^2 - 2nAx + (n+1-1)x^2\\ &= n(n+1)B + n(x^2 - 2Ax + A^2)\\ &= n(n+1)B + n(x-A)^2.\end{aligned}$$

Hence

$$\boxed{D = \frac{n}{n+1}B + \frac{n}{(n+1)^2}(x_{n+1} - A)^2.}$$

**The first inequality.** Multiplying by $n+1$,

$$(n+1)D = nB + \frac{n}{n+1}(x_{n+1} - A)^2 \;\geq\; nB,$$

since the last term is a non-negative multiple of a square. This holds for every value of $x_{n+1}$, with equality exactly when $x_{n+1} = A$.

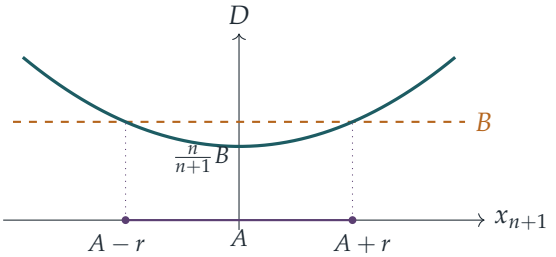
**Ask yourself** How do we turn $D < B$ into a condition on $x_{n+1}$ without worrying about square roots of negative numbers?

**The second inequality.** Note first that $B \geq 0$, being an average of squares, so $\sqrt{(n+1)B/n}$ is a real number. Now, using the boxed formula, each step below is reversible (we only subtract the same quantity from both sides and multiply by the positive number $(n+1)^2/n$):

$$\begin{aligned}D < B &\iff \frac{n}{(n+1)^2}(x_{n+1} - A)^2 < B - \frac{n}{n+1}B = \frac{B}{n+1}\\ &\iff (x_{n+1} - A)^2 < \frac{(n+1)B}{n}\\ &\iff |x_{n+1} - A| < \sqrt{\frac{(n+1)B}{n}}\\ &\iff A - \sqrt{\frac{(n+1)B}{n}} < x_{n+1} < A + \sqrt{\frac{(n+1)B}{n}}.\end{aligned}$$

The third step takes non-negative square roots of both sides, which preserves a strict inequality between non-negative numbers. This is the required "if and only if". (If $B = 0$ the interval is empty, and indeed then $D \geq 0 = B$ always.)

The picture below shows $D$ as a function of $x_{n+1}$ for $n = 3$, $A = 0$, $B = 1$: a parabola with minimum $\frac{n}{n+1}B$ at $x_{n+1} = A$, dipping below the level $B$ exactly on the interval found above.




Here $r = \sqrt{(n+1)B/n}$.

## Checks

- Take $n = 1$: then $B = 0$, and the formula gives $D = \frac{1}{4}(x_2 - x_1)^2$, which is right for the two numbers $x_1, x_2$, whose mean is their midpoint.
- Take $x_1, x_2, x_3 = 0, 0, 3$, so $A = 1$, $B = 2$, and add $x_4 = 1 = A$. Directly, $D = \frac{1}{4}(1 + 1 + 4 + 0) = \frac{3}{2} = \frac{3}{4}B$, the minimum value.
- We checked all three results exactly (with fractions) on 20 000 random lists, including the equivalence $D < B \iff (x_{n+1} - A)^2 < (n+1)B/n$, and confirmed $D = B$ at both endpoints of the interval in 216 cases where they are rational.

## Common slips

- Forgetting that $\sum_{k=1}^{n} A^2 = nA^2$, not $A^2$, when expanding in part (ii).
- Writing $D$ in terms of $\sum x_k^2$ and stopping there: the question wants it in terms of $B$, $A$, $x_{n+1}$ and $n$, so the sum must be replaced using part (ii).
- Square-rooting $(x_{n+1} - A)^2 < k$ to get $x_{n+1} - A < \sqrt{k}$ only, losing the lower end of the interval.

## Beyond STEP

The formula for $D$ is an *updating formula*: it lets a computer maintain the mean and variance of a data stream one value at a time without storing all the data, an idea refined in Welford's algorithm. The term $n(x_{n+1} - A)^2/(n+1)^2$ is a small instance of the decomposition of variance into "within-group" and "between-group" parts that

underlies analysis of variance in statistics. For more on this, see Grimmett and Welsh, *Probability: An Introduction*.

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